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1. Set: private owner key (passkey, Yubikey, Okta, etc)
2. Set: Who has access? (which public keys are valid to read)

Bundle it up:

1. encrypted file
2. What "owner" key is (verify you own the private key)
3. approved public keys to access

Alice

Send via
file transfer

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- Encrypted Bundle
1. encrypted file
 2. owner key
 3. approved public keys

Bob

5

- De-cryption of Bundle if
proof that key pair matches
approved public key list

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Verified Identity
"I have the owner private key"

Yubikey

Okta IdP

FIDO2
Passkey

User Authentication

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Verified Identity
"I have a valid private key that matches an approved public key"

Yubikey

Okta IdP

FIDO2
Passkey

User Authentication

Benefits

- no passwords needed
- encryption is unique to each file
- no servers or central access control, client-only model so offline, airgapped, decentralized, etc
- recipient verifies using their preferred method.

Drawbacks

- owner cannot change permissions. once bundle is sent those public/private keys can access it